

Article

Micro-Scale Agent-Based Modeling of Hurricane Evacuation Under Compound Wind–Surge Hazards: A Case Study of Westbrook, Connecticut

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Abstract

Hurricanes create compound hazards such as storm surge, flooding, and wind-driven debris that can degrade roadway capacity, fragment network connectivity, and hinder evacuation and shelter operations. From a sustainability perspective, improving evacuation planning is essential for reducing disaster-related losses, protecting vulnerable populations, and strengthening the resilience of coastal communities facing intensifying climate-driven hazards. This paper develops a micro-scale, agent-based evacuation modeling framework to assess evacuation performance under baseline and compound-hazard conditions, with emphasis on municipal decision support. The framework is demonstrated for Westbrook, Connecticut, at the census block-group scale in AnyLogic by integrating household locations, vehicle availability, road-network connectivity, and shelter capacities from publicly available datasets. Evacuation propensity and destination choice are parameterized using survey data, enabling empirically grounded decisions for in-town versus out-of-town evacuation among household-vehicle agents. Compound disruptions are represented through flood-related road closures derived from SLOSH storm-surge outputs and stochastic wind-related disruptions that dynamically constrain accessibility during the simulation. Scenarios are evaluated for Saffir–Simpson Category 1–2 and Category 3–4 hurricanes under baseline and compound conditions. Model outputs quantify normalized evacuation time, congestion and critical intersections, shelter demand and unmet capacity, evacuation failure, and spatial heterogeneity across block groups. Results indicate that compound flooding substantially increases evacuation times and failure rates, with the largest performance degradation concentrated in higher-vulnerability areas. Optimization experiments further compare the effectiveness of behavioral shifts, shelter-capacity expansion, and earlier departure timing in reducing delays and unmet shelter demand. Overall, the proposed framework provides transparent, reproducible, and scalable analytics that town engineers and emergency planners can use to evaluate evacuation readiness under compound hurricane impacts.



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1. Introduction

Hurricane evacuations present complex operational challenges driven by the interaction of human behavior, transportation infrastructure, and evolving hazard conditions.

Coastal storms frequently generate compound impacts, including storm surge, inland flooding, and wind-induced debris, which can degrade road networks and restrict access to shelters during evacuation [1–3]. These hazards often evolve rapidly, creating conditions in which evacuation performance is sensitive not only to storm intensity, but also to evacuation timing, destination choice, and the spatial distribution of evacuees across the road network. Transportation congestion, shelter saturation, and cascading infrastructure disruptions can significantly delay clearance and increase the likelihood that some households fail to evacuate entirely [4–6].

Agent-based models (ABMs) have emerged as a widely used approach for simulating evacuation processes owing to their capacity to represent heterogeneous households, nonlinear congestion effects, and feedback between individual decisions and network conditions [7–9]. Prior studies have applied ABMs to hurricanes, tsunamis, floods, and wildfires, demonstrating their capacity to estimate clearance times, characterize congestion patterns, and evaluate shelter utilization under diverse hazard and behavioral assumptions [10–12]. Within hurricane evacuation research, recent ABM applications have increasingly focused on scenario sensitivity and model realism, including destination choice and compliance responses under varying storm intensities and warnings [13–15]. Even as hurricane impacts have grown in recent years, explicit ABM representations of compound hazards and dynamic network accessibility constraints remain scarce.

Several limitations persist in the evacuation modeling literature that restrict its applicability for local planning and decision support. First, evacuation behavior in many ABMs is generated through abstract algorithms, optimization routines, or predictive models rather than being grounded directly in observed evacuation behavior from representative surveys [16,17]. While these approaches can reproduce aggregate patterns, they may obscure how real households make evacuation decisions under different storm conditions. Second, evacuees are often introduced into models at aggregated centroids or arbitrary locations on the road network, rather than entering from realistic residential access points, limiting the ability to analyze neighborhood-scale evacuation performance [18,19]. Third, shelter modeling frequently emphasizes total capacity or optimal shelter placement, without identifying which neighborhoods generate unmet demand or how shelter deficits propagate spatially across a community [20,21]. Fourth, although recent work has begun to incorporate flood impacts into evacuation simulations, relatively few studies explicitly model compound hazards such as storm surge and wind-driven debris together within an ABM framework [1,2]. Finally, socioeconomic characteristics are commonly used as inputs to evacuation decision models but are rarely examined as outcome drivers to assess disparities in evacuation time, shelter access, and evacuation failure across communities [22,23].

This paper addresses these modeling limitations by developing an agent-based evacuation model that emphasizes empirically grounded behavior, spatial realism, and practical applicability for municipal planning. The model integrates a GIS-based road network with dynamic flood-induced road closures derived from storm surge simulations, enabling evaluation of evacuation performance under both baseline and compound-hazard conditions. The framework prioritizes interpretable outputs such as evacuation time, shelter demand, congestion locations, and evacuation failure that can be readily used by town engineers and emergency planners.

Compared to existing evacuation models, the novelty of the proposed framework lies in explicitly linking empirically derived behavioral parameters, fine-scale spatial representation, and dynamic compound-hazard disruptions within a unified agent-based environment. This allows the model to move beyond aggregate clearance estimates and instead produce decision-relevant, neighborhood-scale metrics of evacuation performance, including localized congestion, shelter deficits, and evacuation failure. As

such, the primary scientific contribution of this work is not only in model integration, but in enabling a shift from system-level evaluation to spatially resolved, data-informed evacuation analysis. The evacuation model was implemented using the AnyLogic (v8.9.6) agent-based simulation platform because it enables tight coupling between behavioral decision rules and dynamically evolving transportation networks, allowing explicit representation of compound-hazard-driven road access restrictions during evacuation. A comparison of ABM platforms and their capabilities for such applications is provided in [24]. The model is applied to the Town of Westbrook, Connecticut, to examine evacuation performance across storm categories, identify critical bottlenecks, evaluate behavioral and capacity-based interventions, and assess socioeconomic disparities in evacuation outcomes. Evaluating evacuation performance at neighborhood scale also supports sustainability goals by identifying disparities in evacuation access and infrastructure reliability across different socioeconomic groups, helping planners develop more equitable and resilient evacuation strategies.

The remainder of this paper is organized as follows: Section 2 reviews related evacuation modeling literature; Section 3 describes the agent-based model design and data integration; Section 4 presents the study area and datasets; Section 5 reports simulation results; Section 6 discusses key findings and implications; and Section 7 concludes with limitations and directions for future work.

2. Literature Review

Agent-based models (ABMs) have become a central framework for representing complex evacuation behavior across diverse hazards and built environments. Applications span hurricanes, wildfires, landslides, tsunamis, and chemical accidents [10,25–28], as well as settings ranging from transportation hubs and healthcare facilities to schools, concert venues, industrial sites, and informal settlements [29–34]. Methodological advances have strengthened realism through enhanced behavioral and decision representations, including stress and emotional responses [35,36], detailed motion dynamics [37,38], demographic heterogeneity [39], strategic and game-theoretic decision rules [40,41], and integration with building information models to better capture built-environment constraints [42].

Probabilistic decision rules are increasingly preferred over deterministic formulations in evacuation ABMs because they better capture behavioral uncertainty under evolving emergency conditions [16,43]. For instance, Alqurashi and Altman [44] applied probabilistic choice structures to approximate near-optimal crowd behavior, while [17] modeled evacuation as risk-based decisions conditioned on individual characteristics, capacity constraints, and hazard exposure. Although hurricane-focused ABMs successfully reproduce large-scale traffic patterns and evacuation orders [4,5,13], agent behavior is often derived from algorithms calibrated to socioeconomic variables or machine-learning predictors [45,46] rather than directly anchored to survey-based evacuation rates and destination preferences across surge zones. Evacuees are also frequently introduced at abstracted centroids instead of realistic neighborhood access points. These limitations underscore the need for neighborhood-scale models that empirically ground both behavioral probabilities and household-to-road entry locations.

Beyond behavioral realism, substantial research has advanced the representation of traffic dynamics and clearance estimation in evacuation modeling [47,48]. Dynamic traffic assignment frameworks capture time-varying demand, background traffic, and multimodal interactions more effectively than static clearance calculations [15,49]. ABMs and mesoscopic models further incorporate congestion feedback, rerouting, and spillback [39,50], demonstrating how communication-induced departure surges intensify bottlenecks [51,52] and how locally rational detours can reduce overall system efficiency [53]. Performance

metrics such as total evacuation time, travel speeds, crowd density, and sheltered versus unsheltered populations have consequently become standard [36,54]. Nevertheless, most evidence remains centered on aggregate clearance measures, with limited attention to neighborhood-scale outcomes. This creates a need for block-group-level analyses that evaluate evacuation effectiveness and examine how origin-based destination and shelter splits shape clearance time and success under local network and capacity constraints.

Shelter modeling within evacuation ABMs has advanced considerably, yet it often prioritizes system-wide totals over the spatial distribution of unmet demand. Recent studies incorporate shelter capacity and location [55] and apply multi-objective optimization to enhance evacuation performance and reduce mortality [14]. Others examine how contextual factors such as language accessibility, tourism, and multi-level shelter options influence utilization [56,57], and question the assumption that evacuees choose the nearest facility [58]. Du et al. [21] show that increasing shelter numbers alone does not improve outcomes without strategic siting. Related work identifies imbalanced shelter loading and calls for reassessment of placement and resource allocation [17,20]. Despite these advances, many models still rely on aggregate metrics such as total occupancy or mortality reduction, limiting their ability to trace overflows to specific neighborhoods or assess how targeted capacity adjustments would redistribute deficits. This highlights the need for block-group-level shelter analyses that integrate origin–destination mapping to localize shortages and evaluate practical interventions, such as adding capacity in underserved zones.

Infrastructure failures and compound hazards are gaining attention in evacuation modeling, yet several locally critical processes remain underrepresented. Flood-induced road-loss studies show how inundation can isolate communities and redirect flows, linking exposure to elevation and network structure [59,60]. Related research demonstrates that building collapse and debris increase evacuation times, restrict shelter access, and elevate mortality, emphasizing the need to model obstructions and time-varying network availability [61,62]. Emerging compound-hazard frameworks integrate storm surge, rainfall, and wind damage within broader risk assessments [1,11,14], while network resilience studies reveal cascading disruptions in coupled transport and communication systems as agents adapt and compliance fluctuates [63,64]. Nevertheless, hurricane-focused ABMs rarely represent tree fall or wind-driven debris on local roads, and compound wind–surge impacts are often treated as static disruptions rather than stochastic link failures with recovery dynamics. This underscores the need for neighborhood-scale ABMs that model inundation and debris as endogenous, time-varying processes shaping road availability and evacuation performance.

Behavioral and demographic research provides detailed evidence on evacuation timing and destination choice, yet its integration into ABMs is often one-directional, informing decision rules more than evaluating outcome disparities. Destination choice models include multinomial logit [65], gravity [66], time-dependent discrete choice [67], mean-field games [68], and machine-learning approaches [69], capturing how risk, storm attributes, network conditions, and accommodation availability shape decisions. Empirical studies further examine reliance on public shelters versus private accommodation, highlighting the roles of income, renter status, race, social networks, risk perception, and prior disaster experience [70–73], with recent work offering more granular estimates of shelter and lodging preferences [12,74,75].

Departure-time modeling has similarly moved beyond static utility assumptions to incorporate learning and adaptation [76], participation and information credibility [77], social vulnerability [78], joint destination–departure decisions [79,80], and warning dissemination lags [81], often through hazard-based or joint discrete–continuous formulations [82]. Mamdoohi and Miller-Hooks [83] show that effective information dissemination can re-

duce unnecessary rerouting and accelerate network stabilization. While socioeconomic and vulnerability indicators are widely used to predict evacuation behavior, they are less frequently applied to assess disparities in outcomes such as clearance time, isolation risk, and unmet shelter demand across income, renter, age, disability, or vehicle-access groups.

A parallel discussion concerns modeling scale, simulation platforms, and validation. Microscopic models capture fine-grained movement interactions that influence evacuation time [38,84], whereas mesoscopic approaches emphasize aggregate flows and system-level dynamics [85]. Hybrid frameworks seek to integrate these strengths by linking individual- and density-based representations and connecting micro-level parameters to macro-level outcomes [35]. For instance, Bulumulla et al. [86] combined cognitive, social, and physical submodels in a 35,000-agent simulation, demonstrating that multi-level modeling can substantially alter evacuation performance.

Validation remains challenging, prompting machine-learning-assisted calibration, VR-based behavioral data collection, sequential tuning to address equifinality, and structured multi-step validation protocols [56,87,88]. However, for local emergency managers and municipal engineers, such sophisticated platforms and calibration schemes can appear opaque and difficult to replicate. Results are also not always presented in actionable formats, such as neighborhood-level shelter deficits or block-group clearance-time comparisons needed for planning and mitigation decisions.

Taken together, prior work demonstrates that evacuation ABMs can represent heterogeneous behavior, congestion dynamics, shelter choice, and increasingly complex hazard conditions. Yet several limitations constrain their utility for town-level planning. Survey-reported evacuation rates and destination preferences by surge zone are rarely used to directly parameterize agent decisions, and household-to-network entry points based on buildings and local streets are often simplified. Shelter analyses typically emphasize aggregate occupancy rather than tracing unmet demand to specific neighborhoods. Compound wind and surge impacts are also seldom modeled as stochastic road failures with clearance times, and socioeconomic indicators are more frequently used to generate behavior than to assess disparities in evacuation outcomes. Regional studies in the northeastern United States further highlight evacuation planning challenges specific to coastal New England communities. For example, analyses of evacuation planning and shelter allocation for Aquidneck Island, Rhode Island demonstrate that limited bridge connections and constrained evacuation routes can significantly restrict regional evacuation capacity and require careful coordination of shelter locations and vehicle flows [89,90]. Such localized analyses emphasize the importance of incorporating geographically specific infrastructure constraints and shelter availability when modeling evacuation dynamics in coastal New England towns.

This study addresses these gaps by integrating survey-informed behavior, building- and block-group-based entry points, block-group-level shelter demand and deficit mapping, stochastic treefall and inundation disruptions, and socioeconomic stratification of outcomes within an AnyLogic framework designed to produce decision-ready metrics for local practitioners. In this context, the key contribution of this study is to bridge the gap between methodological advances in evacuation modeling and their practical application for local decision support. By integrating behavior, infrastructure, hazard dynamics, and socioeconomic outcomes within a unified agent-based framework, the model provides actionable insights that are not readily captured by existing approaches focused on aggregate or single-component analyses.

3. Methodology

This study develops a micro-scale agent-based model to simulate hurricane evacuation dynamics based on the framework introduced by [24]. The framework, summarized in Figure 1, integrates evacuation behavior, shelter capacity, road accessibility, and compound hazard effects to evaluate how these factors shape community-level evacuation performance. Agent-based modeling captures household-level decision processes, spatial heterogeneity in vulnerability, and the emergent traffic patterns that arise when many individuals attempt to evacuate simultaneously. By representing vehicles, shelters, and road segments as interacting agents within a GIS-based transportation network, the model reproduces the localized bottlenecks, shelter overcrowding, and route failures that influence overall evacuation outcomes.

Block groups serve as the fundamental spatial unit for population distribution, allowing household and vehicle counts to be assigned with fine-scale accuracy. The model incorporates building footprint data to align households with nearby minor roads, which serve as entry points into the network. This structure reflects how residents realistically access main evacuation corridors, ensuring that agents enter the system from actual locations rather than from aggregated or arbitrary origins.

Evacuation behavior is parameterized using the 2016 New England Household Evacuation Survey—NEHES [91], which reports participation rates and destination patterns by storm-surge zones. Household-vehicle agents follow these survey-based in-town versus out-of-town probabilities, with decisions constrained by shelter capacity, road conditions, and hazard evolution. Vehicle movement is simulated in AnyLogic using its GIS-based road traffic engine on the existing network, with links defined by functional class, speed limit, and lane configuration. The framework incorporates dynamic network degradation by modeling inundation from storm surge and road closures caused by wind-induced tree debris. These disruptions reduce link capacity or render certain segments inaccessible, forcing agents to reroute or, in some cases, prevent evacuation. This enables evaluation of how concurrent flooding and debris affect evacuation time and accessibility.

Shelters are represented as fixed-capacity facilities using FEMA National Shelter System data [92]. Agents evacuating in-town or out-of-town are routed to the nearest shelter with available capacity. When in-town shelters reach capacity, remaining in-town evacuees are redirected to out-of-town shelters along primary evacuation routes. If all shelters are full, agents continue northbound toward inland destinations outside surge-prone areas. This structure quantifies shelter utilization and unmet demand across behavioral and hazard scenarios.

Simulations are conducted for two hazard intensities, Category 1–2 and Category 3–4. Each scenario is run for multiple stochastic iterations to capture variability in agent interactions, debris probability, and shelter arrival sequence. Outputs include evacuation time, shelter deficits, bottleneck locations, and block-group evacuation success rates. Because inputs rely on datasets commonly used by local agencies, including population counts, building footprints, shelter capacity, and survey-based behavior, results are directly interpretable for municipal planners and emergency managers.

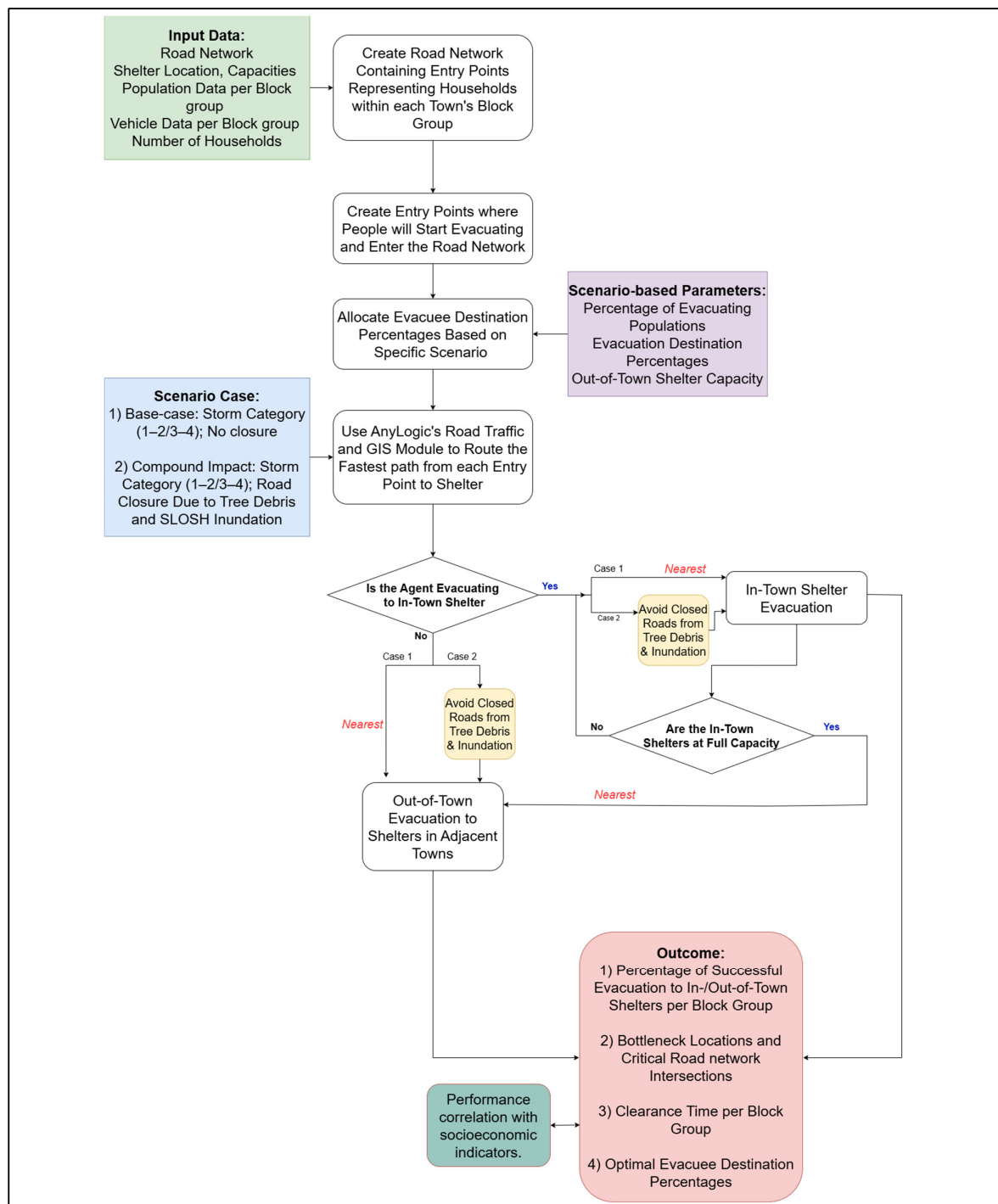


Figure 1. ABM Methodology Flowchart.

3.1. Agent-Based Simulation

This section describes the agent-based model components, including (i) survey-parameterized evacuation behavior, (ii) shelter and destination assignment with capacity constraints, and (iii) the road network with dynamic hazard-induced disruptions.

3.1.1. Evacuation Behavior

Evacuation behavior in the model is grounded in empirically observed patterns rather than synthetic or purely algorithmic decision rules. Each vehicle agent represents a household vehicle available for evacuation, derived from population and household characteristics in the 2023 American Community Survey (ACS) 5-year block-group data [93], and

follows an evacuation decision process informed by the 2016 NEHES. The latter provides surge-zone-specific participation rates, available vehicles for evacuation, and destination tendencies for households in no-surge areas and in SLOSH Zone A and Zone B (Sea, Lake, and Overland Surges from Hurricanes). These parameters are used to assign evacuation probabilities for Category 1–2 and Category 3–4 hurricane scenarios.

During model initialization, each household is assigned an evacuation decision using a probabilistic draw consistent with the survey-derived parameters. Households that evacuate then determine whether to seek refuge at an in-town shelter or travel to an out-of-town destination, based on NEHES-reported in-town versus out-of-town preferences for the corresponding surge zones. Using survey responses directly reduces reliance on abstract behavioral assumptions and reflects reported population tendencies. Departure timing is synchronized across agents and occurs at the start of the simulation, consistent with a mandatory evacuation context. Because the model focuses on system-level clearance and shelter outcomes, late departures and re-entry behavior are not represented.

The number of vehicles assigned to in-town and out-of-town evacuation is computed using Equations (1) and (2), respectively. These expressions determine the number of vehicles N from each block group w that evacuate to in-town or out-of-town shelters under storm category i , SLOSH surge zone j , and town k . In these equations, Z represents the share of households that evacuate for the specified storm category and surge zone, U represents the share of evacuees selecting an in-town or out-of-town destination, and x denotes the share of available vehicles used to evacuate. The terms “*Veh per HH_w*” and “*Pop per HH_w*” capture the average number of vehicles and persons per household in each block group w , while Pop_w is the total population of block group w . These relationships scale NEHES-reported behavioral tendencies to local household conditions, ensuring that vehicle counts reflect both survey behavior and block-group household characteristics.

$$N_{IN(i,j,k,w)} = Z_{IN(i,j,k)} \times U_{IN(i,j,k)} \times x_{(i,j,k)} \times \frac{Veh\ per\ HH_w}{Pop\ per\ HH_w} \times Pop_w \quad (1)$$

$$N_{OUT(i,j,k,w)} = Z_{OUT(i,j,k)} \times U_{OUT(i,j,k)} \times x_{(i,j,k)} \times \frac{Veh\ per\ HH_w}{Pop\ per\ HH_w} \times Pop_w \quad (2)$$

Household location governs network entry. Each household is located using building footprints from the State GIS Office [94] and is linked to the nearest minor road as its network entry point. This places vehicles at realistic neighborhood origins rather than at major roads or block-group centroids. Vehicles then route to their assigned destinations subject to evolving road conditions. This approach preserves household-level spatial realism while remaining consistent with empirically derived evacuation behavior.

3.1.2. Shelter and Destination Modeling

Destination choice and shelter assignment are designed to reflect the options available in the study area. Shelter locations and baseline capacities are obtained from the FEMA National Shelter System [92]. Each vehicle agent is assigned an in-town or out-of-town destination at initialization using NEHES-derived probabilities by surge zone and storm category [91]. For in-town evacuees, agents are routed to the nearest available shelter. Shelters operate under fixed capacities; when a shelter reaches capacity, arriving vehicles are redirected to the next nearest shelter until all local shelters are full.

Agents assigned to out-of-town evacuation travel along major corridors connecting the study area to neighboring inland communities. Out-of-town shelters are modeled as fixed-capacity facilities in adjacent towns. Their available capacity accounts for local demand from evacuees originating within those towns. Accordingly, the available capacity

for an out-of-town shelter in town k , under storm category i and SLOSH zone j , is calculated using Equation (3):

$$C_{OTS(i,j,k),available} = C_{OTS_k} - D_{(i,j,k)} \quad (3)$$

where C_{OTS_k} is the NSS-reported shelter capacity and $D_{(i,j,k)}$ is the estimated demand from evacuees within that town. The local demand term $D_{(i,j,k)}$ is computed using Equation (4), which applies the same structure used for calculating in-town evacuees in Equation (1).

$$D_{(i,j,k)} = Z_{IN(i,j,k)} \times U_{IN(i,j,k)} \times x_{(i,j,k)} \times \frac{Veh\ per\ HH_k}{Pop\ per\ HH_k} \times Pop_k \quad (4)$$

where Z represents the percentage of people evacuating in town k under storm category i and SLOSH zone j , U_{IN} is the proportion evacuating to in-town shelters, and x is the proportion of vehicles available for evacuation. The demographic scaling terms “ $Veh\ per\ HH_k$,” “ $Pop\ per\ HH_k$,” and Pop_k ensure that demand reflects the local household and vehicle characteristics [93].

Once both in-town and out-of-town shelters reach capacity, vehicles continue traveling northbound along the primary evacuation route, representing households that leave the region when formal sheltering options are unavailable. This structure enables the model to track shelter occupancy and overflow and to identify the block groups contributing to excess demand. By combining household-level entry points with capacity-constrained shelters, the model captures how shelter access varies across the community under different hazard and behavioral scenarios.

3.1.3. Road Network

The road network is parameterized using roadway geometry and functional classification from the U.S. Department of Transportation Highway Performance Monitoring System (HPMS) [95], which provides segment-level attributes such as functional class, number of lanes, speed limits, and surface characteristics. These data are imported into AnyLogic’s GIS environment to represent local streets, collectors, primary evacuation corridors, and connections for out-of-town travel. Vehicle movement follows AnyLogic’s queueing and lane-changing logic, enabling congestion formation, spillback, merging dynamics, and travel-time variability. Agents enter the network from realistic household access points by linking residential structures to the nearest minor road segment. Network availability updates during runtime as links experience capacity reductions or closures from storm surge inundation and wind-driven debris (Section 3.3); vehicles reroute based on the updated topology, and agents with no feasible path are classified as unable to evacuate. Overall, the model combines HPMS-based roadway attributes with dynamic hazard-induced disruptions to capture how traffic operations and evolving accessibility jointly shape evacuation performance.

3.2. Evacuation Scenarios

The model evaluates evacuation performance under hurricane scenarios that vary by storm intensity and whether hazard-induced road disruptions are present. Two baseline scenarios represent Category 1–2 and Category 3–4 storms, with evacuation participation rates and destination distributions derived from the 2016 NEHES dataset. Storm intensity follows the Saffir–Simpson Hurricane Wind Scale, a classification based on maximum sustained wind speed; Category 3 and higher are major hurricanes, with sustained winds of at least 129 mph [96,97]. Baseline simulations assume all road segments remain open, establishing reference clearance times, shelter demand, and traffic patterns under ideal network conditions.

The second set of scenarios represents compound hazards by introducing storm-surge inundation and wind-induced tree debris that alter network availability. Road closures

follow the depth thresholds and debris probabilities described in Section 3.3, producing link failures that reduce connectivity and can prevent some block groups from reaching shelters or exiting the town. These scenarios quantify how compound hazards can affect evacuation time, shelter overflow, and the share of households unable to evacuate.

For each storm category, baseline and compound-hazard scenarios are simulated for 100 stochastic iterations to capture variability in traffic interactions, shelter arrival sequences, and debris-induced closures. Ensemble outputs are summarized using means and percentile ranges to provide stable evacuation performance estimates and support comparisons across storm intensities and hazard conditions while accounting for uncertainty in both population movement and infrastructure disruption.

3.3. Flooding and Wind-Induced Tree Debris Effects

Hurricane evacuations are often disrupted by simultaneous hydrologic and wind-driven impacts that reduce roadway accessibility and fragment transportation networks. To capture these compound effects, the model incorporates storm surge inundation and wind-induced tree debris as hazard-driven constraints on road availability and evacuation performance.

3.3.1. SLOSH Model

Flooding effects from storm surge were incorporated using outputs from NOAA's SLOSH model version 4 [98] which are based on the map products released in 2014 [99]. SLOSH provides raster estimates of maximum storm-surge inundation for multiple hurricane intensities. In this study, Category 2 and Category 4 rasters were used to represent surge conditions for the Category 1–2 and Category 3–4 evacuation scenarios, respectively. These rasters provide spatially continuous water depth estimates to identify road segments exposed to inundation.

SLOSH rasters were processed in ArcGIS Pro 3.4.3 to delineate impassable roads. For each storm category, the inundation raster was overlaid with the roadway network, and segments intersecting depths of 1 ft (0.305 m) or greater were classified as closed. This threshold is consistent with prior findings on vehicle instability and loss of traction in floodwater, including critical-depth estimates for passenger vehicles [100] and buoyancy-related risks at shallow depths [101]. The resulting closed-road set was exported and used to modify the evacuation network in AnyLogic.

Road closures were implemented in AnyLogic using the "Avoid Road" functionality, enabling agents to reroute around inundated segments. Roads flagged in ArcGIS Pro were assigned to the avoidance list before each simulation. Because SLOSH represents maximum inundation extent rather than time-varying water-depth progression, closures were held fixed for the duration of each scenario. This integration translates physically based storm-surge estimates into network accessibility constraints and captures the connectivity loss expected during flooding events.

3.3.2. Wind Model

To represent tree-debris road blockages, datasets on the transportation network and forest structure were compiled. Roadway centerlines were obtained from the U.S. DOT HPMS [95]. A 3 m buffer was applied to account for roadway width. Tree height was characterized using the Global Forest Canopy Height—GLAD product [102]. The dataset integrates NASA's Global Ecosystem Dynamics Investigation (GEDI) lidar forest structure measurements with Landsat data time-series to provide 30 m resolution canopy height estimates [102]. Although machine-learning approaches exist for automated tree detection [103–105], they typically require extensive labeled training data specific to the study

area. Accordingly, tree points were digitized manually through visual interpretation using 0.3 m resolution imagery from the National Agricultural Imagery Program [106].

Wind speed is required at each tree point to evaluate wind loading on tree structures. The Weather Research and Forecasting (WRF) model [107] was used to simulate wind fields for six major historical windstorms, including Hurricane Carol (1954), Hurricane Gloria (1985), Hurricane Bob (1991), Hurricane Irene (2011), Hurricane Sandy (2012), and Tropical Storm Isaias (2020), at 4 km resolution. For each tree location, hourly maximum gust speed (MaxGust) and wind direction were extracted over a two-day window. To represent a conservative loading condition, the maximum MaxGust value over the full period was selected for subsequent wind-load evaluation.

Tree allometry and fiber strength vary by species. Thus, forest cover and species distribution were compiled. Following [104], diameter at breast height (DBH), crown height, and crown diameter were estimated using empirically derived allometric relationships [108], and wind response was predicted using finite-element analysis. Tree failure probabilities were then translated into roadway blockage probabilities through proximity analysis. As illustrated in Figure 2, a tree falling along the gust direction may block traffic if its height and distance to the roadway edge are sufficient; such trees are classified as risk-trees, whereas trees located farther from the roadway are assumed unlikely to cause blockage. Proximity calculations were implemented using trigonometric relationships consistent with prior studies [109].

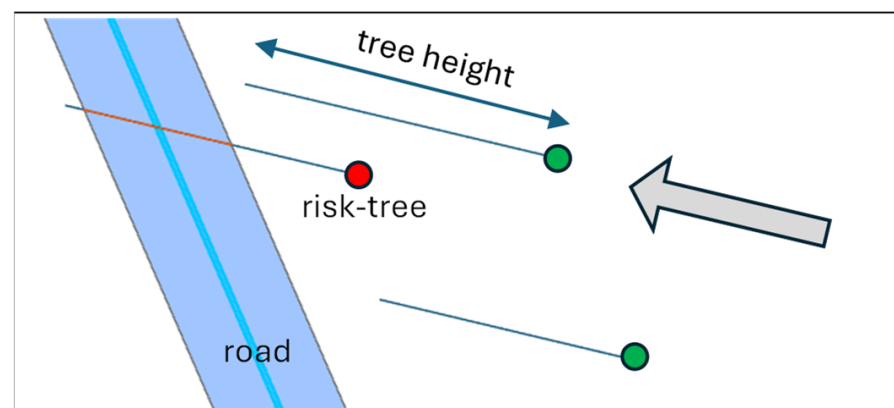


Figure 2. Proximity Analysis for Roadway Blockage Probability.

To support debris-related disruption modeling, forest cover and species composition were compiled because tree allometry and wood strength are species dependent. In Connecticut, oak (62.9%), maple (20.4%), and ash (9.5%) comprise 92.8% of forest cover [110], informing the representation of tree characteristics relevant to wind-driven failure and potential roadway blockage.

Because blockage occurs if at least one associated tree fails, the blockage probability of n th road segment, P_n , is modeled using Equation (5) as a series system of independent tree failures:

$$P_n = 1 - \prod_m (1 - p_{n,m}) \quad (5)$$

where $p_{n,m}$ is the failure probability of m th tree associated with the n th segment.

The lognormal fragility curves for the three species across tree-height classes are shown in Figure 3. For a given wind speed, maple exhibits higher failure probabilities, whereas oak shows the greatest resistance to wind loading. Across all three species, failure probability decreases with increasing tree height, indicating that the increase in diameter at breast height (via allometric scaling) enhances structural resistance more than the concurrent increase in wind load associated with a larger projected area. In addition, oak appears more

sensitive to height than maple and ash, with larger shifts in failure probability across height classes. Using the species- and height-specific fragility curves together with the series-system formulation, road blockage probabilities were computed for each of the six historical windstorms (Hurricane Carol, 1954; Hurricane Gloria, 1985; Hurricane Bob, 1991; Hurricane Irene, 2011; Hurricane Sandy, 2012; and Tropical Storm Isaias, 2020). Among these events, Hurricane Irene produces the highest modeled blockage risk in the study area, which likely reflects a combination of stronger local gust exposure and wind-direction alignment that increases the likelihood that road-adjacent trees fall across travel lanes.

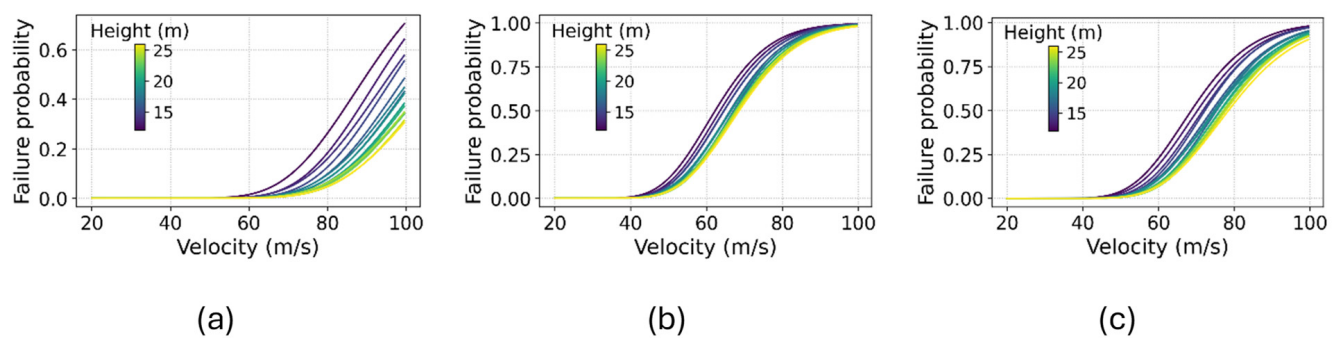


Figure 3. Fragility Curves of Foliated Trees with Different Heights. (a) Oak Tree, (b) Maple Tree, (c) Ash Tree [104].

3.4. Evacuation Performance Metrics

The model generates evacuation performance metrics that quantify how effectively households reach safety under different hazard and behavioral scenarios. Metrics are computed at the block-group level to capture spatial heterogeneity across the study area.

First, evacuation time is measured using the simulation clock and defined as the maximum time required for all successfully evacuating vehicles from a block group to reach an assigned shelter or to exit the town toward inland destinations. Evacuation time is computed separately for each block group to identify spatial differences driven by network structure, shelter accessibility, and local congestion. To account for scenarios in which a portion of vehicles cannot evacuate as intended, the model also reports a normalized evacuation time for each block group, calculated as the block-group evacuation time divided by the share of vehicles that successfully evacuate from that block group. This normalization reduces bias in high-disruption scenarios where failures or rerouting to inland exits may shorten observed travel times and yield deceptively low clearance estimates. The normalized evacuation time for block group w is defined in Equation (6):

$$T_{evac_w} = \frac{\max(Evacuation\ Time_w)}{\%Evacuated\ Vehicles_w} \quad (6)$$

Second, the percentage of successful evacuation is calculated separately for in-town and out-of-town sheltering. For each storm category i and block group w , the in-town success rate is computed as the number of vehicles that reached an in-town shelter, V_{IN} , divided by the number of vehicles that attempted in-town evacuation, N_{IN} . The out-of-town success rate is defined analogously using the number of vehicles that reached out-of-town shelters, V_{OUT} , and the number that attempted out-of-town evacuation, N_{OUT} . These relationships are given in Equations (7) and (8):

$$Evac_{IN(i,w)} = \frac{V_{IN(i,w)}}{N_{IN(i,w)}} \times 100\% \quad (7)$$

$$Evac_{OUT(i,w)} = \frac{V_{OUT(i,w)}}{N_{OUT(i,w)}} \times 100\% \quad (8)$$

Shelter performance is represented through occupancy, peak load, and unmet demand. Each shelter records arrivals, remaining capacity, and the number of vehicles redirected once capacity is reached, enabling estimation of total deficits and the block-group origins of excess demand. Network bottlenecks are identified by tracking queue buildup, reduced vehicle speeds, and recurrent congestion on specific road segments. Vehicles that do not reach any shelter and instead leave the network, or cannot access an inland exit due to closures, congestion, or shelter saturation, are classified as failed evacuations. The failed-evacuation percentage for storm category i and block group w is calculated as the number of vehicles that exit the network without reaching a shelter, V_{EXIT} , divided by the total number of vehicles attempting evacuation, $N_{IN} + N_{OUT}$, as shown in Equation (9):

$$Evac_{FAIL(i,w)} = \frac{V_{EXIT(i,w)}}{N_{OUT(i,w)} + N_{IN(i,w)}} \times 100\% \quad (9)$$

Socioeconomic characteristics are incorporated by linking block-group evacuation outcomes to demographic and vulnerability variables from the 2023 ACS [93]. Socioeconomic pillars include race and ethnicity, population structure, socioeconomic status, housing structure, special needs, and employment structure, consistent with indicators used in social vulnerability assessments such as the Social Vulnerability Index [111,112]. After each scenario, evacuation times and success rates are compared across these demographic strata to assess how performance varies with underlying socioeconomic conditions. Performance metrics were summarized across stochastic iterations using means and percentile ranges to support stable and reproducible comparisons.

3.5. Optimization, Sensitivity, and Validation

The model includes a set of scenario-based, optimization-focused experiments to evaluate how targeted interventions can improve evacuation performance. These experiments do not implement algorithmic optimization; instead, they test practical modifications to shelter capacity and behavioral assumptions and quantify impacts on clearance time, shelter deficits, congestion, and evacuation success.

The first experiment evaluates shelter-capacity expansion in areas with persistent unmet demand under baseline conditions. Block groups with the highest unmet shelter demand are identified from the simulation outputs, and a new in-town shelter with capacity comparable to existing shelters is added within the selected block group at an accessible location on the local road network. Simulations are then repeated under the same hazard and behavioral conditions to measure changes in shelter occupancy, travel patterns, and the percentage of households that evacuate successfully. This approach provides a direct measure of how strategic shelter placement can alleviate congestion and reduce localized evacuation failures.

A second experiment tests behavioral adjustments by varying the in-town versus out-of-town destination shares relative to the NEHES-based baseline. Alternative destination splits are evaluated to assess whether shifts in destination preferences reduce clearance time, reduce bottlenecks near major intersections, or mitigate shelter overflow. For each modified behavioral distribution, the model computes changes in clearance times, successful evacuation rates, and the spatial distribution of unmet demand. This analysis identifies whether certain destination patterns lead to more efficient system-wide performance.

Although a formal global sensitivity analysis was not conducted, the model structure and scenario design enable an implicit assessment of the relative influence of key components on evacuation outcomes. Comparisons between baseline and compound hazard scenarios help isolate the effects of hazard-induced network disruption, including road closures caused by inundation and wind-driven debris, on evacuation time and

failure rates. In addition, scenario-based modifications, such as optimized destination choice and shelter capacity adjustments, provide insight into how behavioral decision rules and infrastructure constraints affect congestion patterns, shelter utilization, and localized evacuation success. By systematically varying these elements across simulation scenarios, the framework supports interpretation of the relative importance of network availability, agent behavior, and shelter capacity in shaping system-level and neighborhood-scale evacuation performance.

Model validation followed an iterative process to assess its accuracy, internal consistency, and alignment with real-world evacuation conditions. Internal validation verified that agent decision rules, shelter assignment, and traffic-flow dynamics operated as intended. Several techniques were applied, including: (i) component testing, in which selected behavioral or routing modules were temporarily removed to evaluate their influence on system outcomes; (ii) extreme-value testing, where evacuation participation, shelter capacities, and road availability were set to upper and lower bounds to confirm logical behavior under controlled conditions; and (iii) statistical validation using repeated random-seed replications, with sampling and bootstrapping to assess stability and variability across stochastic realizations. External validation evaluated whether simulated clearance times, shelter utilization, and congestion patterns were consistent with established trends from past hurricane evacuations in coastal New England and whether modeled behavior aligned with tendencies reported in the 2016 New England Household Evacuation Survey—NEHES. Collectively, these steps support the credibility and interpretability of the model for scenario evaluation and decision-support applications.

4. Study Area

The Town of Westbrook, Connecticut, located on the central shoreline of Long Island Sound, serves as the study area for the evacuation modeling framework. Westbrook is a small coastal municipality in Middlesex County with a total area of approximately 21.4 square miles, including about 15.7 square miles of land and 5.7 square miles of water distributed across five census block groups [93]. The 2023 U.S. Census population estimate is 6698 [93]. The Town is well suited for hurricane evacuation analysis given its exposure to compound coastal hazards, including storm surge and tidal flooding, and the potential for constrained inland routing. Several low-lying neighborhoods fall within SLOSH-designated surge zones, where hurricane inundation can disrupt both major and minor road corridors. Westbrook is bordered by Old Saybrook, Essex, Deep River, and Clinton; the model considers the nearest shelters in these adjacent towns. In total, 11 shelters are represented, including three in Westbrook and eight in surrounding towns. Figure 4 presents the study area, showing block-group boundaries, the road network, shelter locations, and the SLOSH zones used to parameterize evacuation behavior.

Westbrook's transportation system includes major regional corridors and a dense local street network, with several narrow connections and tidal river crossings that can amplify bottlenecks during evacuation. Interstate 95 bisects the town and functions as a primary coastal evacuation route, with Annual Average Daily Traffic (AADT) reaching approximately 72,800 vehicles on segments within Westbrook [95]. U.S. Route 1 also traverses the town and carries up to about 35,800 vehicles per day, serving as a key commercial and residential thoroughfare. Additional regional connectivity is provided by state routes, including CT 9, 81, 145, 153, 154, and 166. A substantial share of Westbrook's local and minor roads lies in low-lying coastal areas and are therefore susceptible to storm-surge inundation. Spatial variation in building density across block groups further contributes to differences in population exposure, available vehicles, and shelter demand.

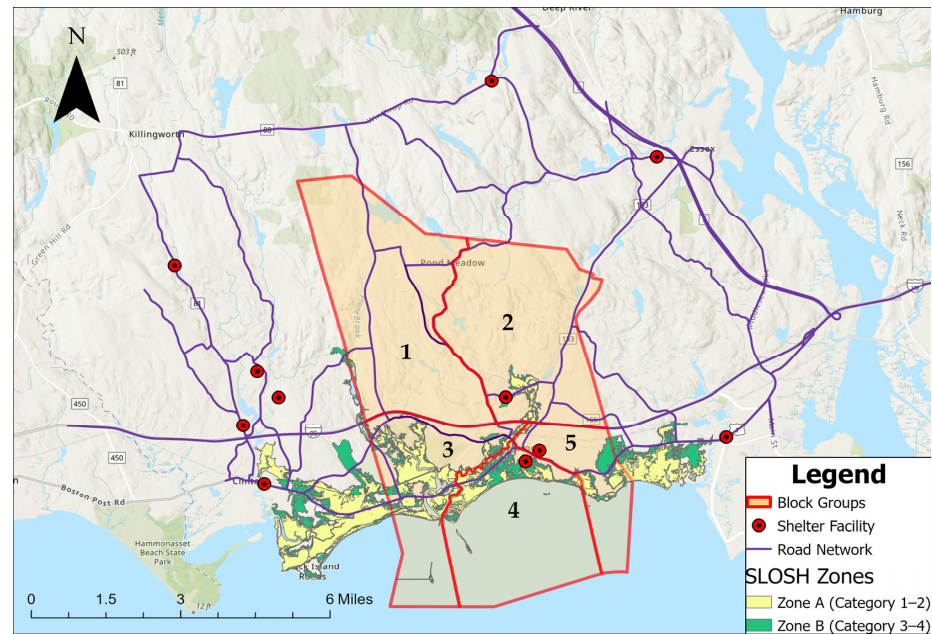


Figure 4. Study Area of Westbrook, Connecticut.

Table 1 summarizes the block-group parameters used in Section 3 and classifies them into three categories. Fixed parameters include population, housing units, vehicle availability, SLOSH-zone designation, and shelter counts and capacities, which define baseline household and infrastructure characteristics. SLOSH-zone-specific parameters include available vehicles used to evacuate and the shares selecting in-town versus out-of-town destinations, derived from the 2016 NEHES and assigned based on each block group's surge-zone classification. Scenario-specific parameters include evacuation participation rates for Category 1–2 and Category 3–4 storms and adjusted out-of-town shelter capacities, which determine the number of agents entering the network under each hazard scenario and the capacity available in adjacent towns after accounting for their local shelter demand.

Table 1. Block-Group Model Inputs and Scenario Parameters.

Parameter	Block Group 1	Block Group 2	Block Group 3	Block Group 4	Block Group 5	Source
Fixed Values						
Population in Housing Units	892	1973	1381	856	1596	[93]
Number of Houses	525	752	996	918	765	[94]
Number of Vehicles	862	1929	1361	852	1215	[93]
SLOSH Zone	Non-surge	Non-surge	Zone B	Zone B	Zone B	[98]
Number of Shelters	-	-	1	1	1	[92]
Shelter Capacity	-	-	1000	300	90	[92]
SLOSH Zone-specific Values						
Available Vehicles to Evacuate in	73%	73%	83%	83%	83%	[91]
Percentage of In-Town Evacuees	33%	33%	42%	42%	42%	[91]
Percentage of Out-of-Town Evacuees	53%	53%	54%	54%	54%	[91]

Table 1. Cont.

Parameter	Block Group 1	Block Group 2	Block Group 3	Block Group 4	Block Group 5	Source
Scenario-specific Values: Storm Category 1–2						
Evacuation Participation	61%	61%	65%	65%	65%	[91]
Out-of-Town Shelters Capacity			Old Saybrook: 51, Deep River: 963, Clinton: 1792, Essex: 535 <i>Total = 3341</i>			[91,92]
Scenario-specific Values: Storm Category 3–4						
Evacuation Participation	74%	74%	77%	77%	77%	[91]
Out-of-Town Shelters Capacity			Old Saybrook: 10, Deep River: 896, Clinton: 1234, Essex: 435 <i>Total = 2724</i>			[91,92]

5. Results & Analysis

This section presents evacuation simulation results for the Town of Westbrook under multiple hurricane scenarios. Results are organized around system-level performance, socioeconomic variation in outcomes, and the effects of targeted intervention experiments. Metrics are reported for baseline and compound-hazard scenarios for Category 1–2 and Category 3–4 storms, with emphasis on evacuation time, destination-specific success (i.e., in-town versus out-of-town sheltering), evacuation failure, congestion patterns, and block-group-level outcomes.

5.1. Evacuation Performance

Figure 5 summarizes evacuation performance across the five block groups for four scenarios: Category 1–2 baseline, Category 3–4 baseline, Category 1–2 compound wind–flooding, and Category 3–4 compound wind–flooding. Evacuation time shows substantial spatial and scenario-based variation. Under baseline conditions, evacuation times are generally 2.8–4.3 h across most block groups for Category 1–2 storms, with higher values under Category 3–4 storms, particularly in Block Groups 4 and 5 (4.3–6.3 h). Compound hazards increase evacuation times in specific locations. The largest increase occurs in Block Group 3 under the Category 3–4 compound scenario, where evacuation time rises to approximately 19.6 h; Block Groups 4 and 5 also exceed 8.0 h and 7.5 h, respectively. In contrast, Block Groups 1 and 2 show minimal sensitivity to compound hazards, with evacuation times slightly decreasing (~3.0 to 2.6 h and 3.05 to 2.6 h) and remaining below 4.5 h across all scenarios.

Destination-specific evacuation outcomes in Figure 5 reveal systematic differences across block groups. In-town evacuation success is consistently highest in Block Groups 1 and 2, reaching 100% in the compound scenarios for both storm categories. Block Groups 3–5 exhibit lower in-town success, with pronounced reductions under compound hazards; for example, Block Group 3 declines from 35% to approximately 11% for Category 3–4 storms when compound disruptions are introduced. Out-of-town evacuation success is generally high in baseline scenarios, often approaching 100%, but deteriorates under compound hazards, including a complete loss of successful out-of-town evacuation in Block Groups 4 and 5 during the Category 3–4 compound scenario. These patterns are mirrored in evacuation failure rates, which are near zero under baseline conditions for

most block groups but rise sharply under compound hazards. In the Category 3–4 compound scenario, failure exceeds 90% in Block Groups 3–5, whereas Block Groups 1 and 2 maintain a 0% failure rate. Overall, the results indicate that compound wind–flooding disruptions disproportionately degrade evacuation performance in Block Groups 3–5 by simultaneously increasing travel times and limiting access to both in-town and out-of-town shelter destinations.

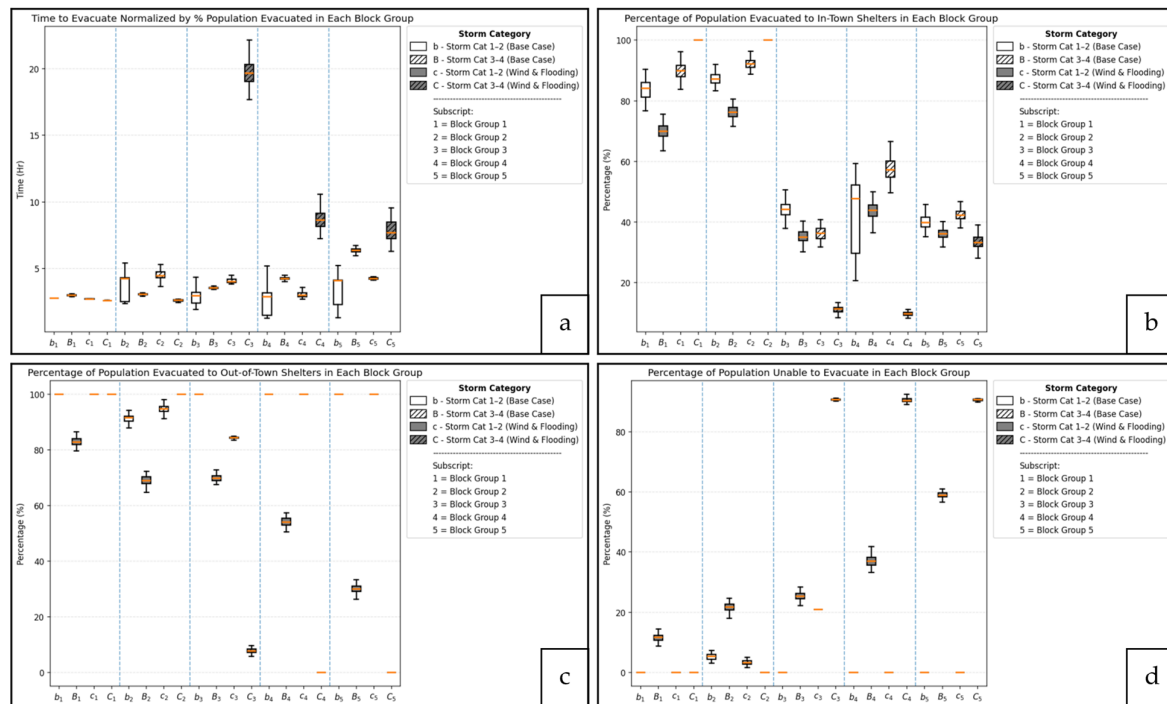


Figure 5. Evacuation Performance by Block Group across Storm Scenarios. (a) Normalized Evacuation Time, (b) In-Town Evacuation Success Rate, (c) Out-of-Town Evacuation Success Rate, and (d) Evacuation Failure Rate, for Category 1–2 and Category 3–4 Storms Under Baseline and Compound Wind–Flooding Conditions. Dashed Blue Lines Indicate Boundaries between Block Groups 1–5.

The comparison of evacuation times across baseline and compound scenarios provides a system-level benchmark for evacuation timing in Westbrook. Without intervention, the baseline scenarios for Category 1–2 and Category 3–4 storms represent the minimum clearance duration under unconstrained network conditions. As shown in Figure 6, the maximum block-group normalized evacuation time in each baseline scenario corresponds to the last block group to clear and therefore defines the effective town-wide clearance time. Under compound wind–flooding conditions, clearance times increase substantially, with the maximum evacuation time reaching approximately 19 h for the Category 3–4 compound scenario, reflecting the combined effects of road closures, rerouting, shelter constraints, and network disruption.

Figure 6 also shows that evacuation times increase for all block groups under compound conditions relative to the baseline, and the difference between these values represents a practical time penalty associated with evacuating later in the event timeline. If evacuation orders are issued sufficiently early so that evacuation is completed within the baseline clearance time (x hours), evacuees can clear before compound impacts degrade accessibility (e.g., inundation-related closures, debris blockages, islanding, etc.). In effect, completing evacuation within x hours prevents escalation toward the compound clearance time (y hours) and reduces both delay and failure. For example, the town-wide clearance time increases from approximately 6.3 h under Category 3–4 baseline conditions to nearly 19 h under Category 3–4 compound conditions, while evacuation failure in high-vulnerability

block groups increases from approximately 25–37% to more than 90% (Figure 5a,d). These results suggest that issuing evacuation orders at least x hours prior to landfall could maintain baseline performance and substantially reduce evacuation failure across the most affected block groups.

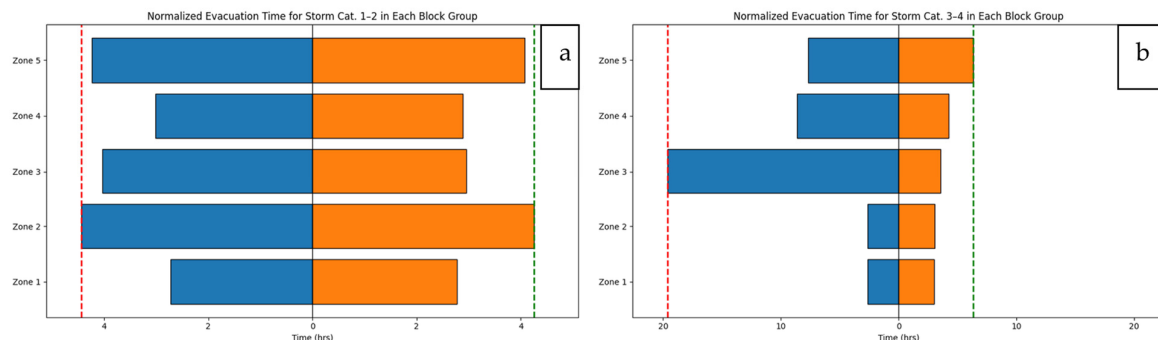


Figure 6. Comparison of Town-Wide Evacuation Clearance Times under Baseline and Compound Flooding Conditions: (a) Storm Category 1–2, and (b) Storm Category 3–4. The green dashed line indicates the maximum evacuation time across block groups under baseline conditions (x -hours), while the red dashed line indicates the corresponding town-wide clearance time under compound flooding conditions (y -hours).

Table 2 summarizes congestion patterns and the emergence of critical intersections across scenarios. Scenario simulations indicate that both the spatial extent and intensity of congestion increase with storm severity and compound disruptions, with a larger share of roadway segments transitioning from 60 km/h to 10 km/h under compound conditions. Consistent with these patterns, the number of identified critical intersections increases from five in the Category 1–2 baseline scenario to eight in each compound-hazard scenario. Several intersections recur across all scenarios, including US Route 1 at Route 153, Route 153 at McVeagh Road, and Pond Meadow Road at McVeagh Road. Block Group 2 contains the highest number of critical intersections in all four scenarios making it the most critical block group in terms of congestion, while Block Groups 3 and 4 also exhibit repeated congestion points under higher-intensity and compound conditions. Additional critical intersections emerge outside the town boundary in adjacent towns, including Essex, Clinton, and Deep River, particularly under compound wind–flooding scenarios, indicating that congestion propagates beyond the immediate study area.

Table 2. Critical Intersections and Congestion Hotspots across Evacuation Scenarios.

Storm Scenario	Critical Intersections (CI)/Roads	Block Groups Affected	Block Group with Highest CIs
Category 1–2 (Baseline)	(1) US Route 1 at US Route 153; (2) US Route 153 at McVeagh Rd; (3) US Route 1 at Liberty St; (4) US Route 153 at Bokum Rd; (5) Pond Meadow Rd at McVeagh Rd	Block Groups: 2 & 4 Out of Town: Essex, Clinton	Block Group 2
Category 1–2 (Wind–Flooding)	(1) US Route 1 at US Route 153; (2) US Route 153 at McVeagh Rd; (3) US Route 1 at Liberty St; (4) US Route 153 at Bokum Rd; (5) Pond Meadow Rd at McVeagh Rd; (6) Main St at Main St No. 2; (7) US Route 153 at Monahan Rd; (8) US Route 153 at US Route 166	Block Groups: 2, 3, & 4 Out of Town: Essex, Clinton, Deep River	Block Group 2

Table 2. Cont.

Storm Scenario	Critical Intersections (CI)/Roads	Block Groups Affected	Block Group with Highest CIs
Category 3–4 (Baseline)	(1) US Route 1 at US Route 153; (2) US Route 153 at McVeagh Rd; (3) US Route 1 at Liberty St; (4) US Route 153 at Bokum Rd; (5) Pond Meadow Rd at McVeagh Rd; (6) US Route 1 at Grove Beach Rd N; (7) US Route 1 at US Route 154	Block Groups: 2, 3, & 4 Out of Town: Essex, Clinton	Block Group 2
Category 3–4 (Wind–Flooding)	(1) US Route 1 at US Route 153; (2) US Route 153 at McVeagh Rd; (3) US Route 153 at Bokum Rd; (4) Pond Meadow Rd at McVeagh Rd; (5) Main St at Main St No. 2; (6) US Route 153 at Monahan Rd; (7) US Route 153 at US Route 166; (8) Dennison Rd at East Pond Meadow Rd	Block Groups: 2, 3, & 4 Out of Town: Essex, Clinton, Deep River	Block Group 2

5.2. Performance Correlation with Socioeconomic Indicators

Table 3 summarizes the significant Pearson correlations between block-group socioeconomic indicators and the four evacuation outcomes for Category 1–2 and Category 3–4 storms under baseline and compound-hazard conditions, respectively.

Table 3. Significant Correlations between Socioeconomic Indicators and Evacuation Performance Metrics under Baseline Conditions.

SoVI Variable	Storm Category 1–2				Storm Category 3–4			
	Evac. Time	In-Town	Out-Town	Evac. Failure	Evac Time	In-Town	Out-Town	Evac. Failure
Baseline Conditions								
QASIAN	0.34	0.97 **	−0.71	0.71	−0.57	0.97 **	0.54	−0.6
QFAM	−0.68	−0.64	1.00 ***	−1.00 ***	0.4	−0.69	−0.22	0.28
QFEMALE	−0.21	0.48	−0.49	0.49	−0.89 *	0.42	0.8	−0.8
QNOHLTH	0.5	−0.4	0.27	−0.27	0.88 *	−0.37	−0.78	0.79
QESL	0.54	−0.5	0.25	−0.25	0.93 *	−0.46	−0.86	0.86
QED12LESS	0.68	0.64	−1.00 ***	1.00 ***	−0.4	0.69	0.22	−0.28
QRENTER	0.37	−0.61	0.28	−0.28	0.93 *	−0.52	−0.96 *	0.94 *
Compound Wind and Flooding Conditions								
QASIAN	0.02	0.94 *	0.23	−0.36	−0.77	0.99 **	0.98 **	−0.98 **
QSSBEN	0.63	−0.4	−0.94 *	0.91 *	0.7	−0.18	−0.11	0.16
QUNEMPL	0.41	−0.83	−0.83	0.89 *	0.97 **	−0.71	−0.64	0.69

* Correlation is slightly significant (two-tailed $p < 0.05$), ** Correlation is significant (two-tailed $p < 0.01$), *** Correlation is highly significant (two-tailed $p < 0.001$).

Under baseline conditions, the most consistent association is for QASIAN (percent Asian), which is positively correlated with in-town evacuation success in both storm categories ($r = 0.97$, $p < 0.01$), suggesting systematically higher local shelter success in block groups with higher Asian population shares. In the Category 1–2 baseline scenario, out-of-town evacuation success is perfectly positively correlated with QFAM (percent of children living in two-parent families; $r = 1.00$, $p < 0.001$) and perfectly negatively correlated with

QED12LESS (percent of the population over age 25 with less than 12th-grade education; $r = -1.00$, $p < 0.001$). Correspondingly, evacuation failure exhibits equal-magnitude correlations in the opposite direction ($r = -1.00$ for QFAM and $r = 1.00$ for QED12LESS), indicating that higher QFAM is associated with higher out-of-town success and lower failure, whereas higher QED12LESS is associated with lower out-of-town success and higher failure among attempted evacuations. For the Category 3–4 baseline scenario, evacuation time is significantly correlated with several socio-demographic and access indicators, including QFEMALE (percent female; $r = -0.89$, $p < 0.05$), QNOHLTH (percent without health insurance; $r = 0.88$, $p < 0.05$), and QESL (percent of households with limited English proficiency; $r = 0.93$, $p < 0.05$), indicating that longer evacuation times are associated with higher uninsured and limited-English-proficiency shares, whereas higher female share is associated with shorter evacuation times. QRENTER (percent renter-occupied households) is also significant in Category 3–4 baseline conditions, correlating with evacuation time ($r = 0.93$, $p < 0.05$), out-of-town evacuation success ($r = -0.96$, $p < 0.05$), and evacuation failure ($r = 0.94$, $p < 0.05$), signifying that renter-dominated block groups experience longer clearance times and lower out-of-town shelter success alongside higher overall failure.

Under compound wind–flooding conditions, several socioeconomic indicators exhibit statistically significant associations with evacuation outcomes, particularly for Category 3–4 storms. QASIAN is strongly positively correlated with in-town evacuation success ($r = 0.99$, $p < 0.01$) and out-of-town evacuation success ($r = 0.98$, $p < 0.01$), and strongly negatively correlated with evacuation failure ($r = -0.98$, $p < 0.01$). In Category 1–2 compound conditions, QSSBEN (percent of households receiving Social Security benefits) is negatively associated with out-of-town evacuation success ($r = -0.94$, $p < 0.05$) and positively associated with evacuation failure ($r = 0.91$, $p < 0.05$), indicating reduced out-of-town shelter success and higher failure in block groups with higher Social Security receipt. QUNEMPL (civilian unemployment) is positively correlated with evacuation failure in Category 1–2 compound conditions ($r = 0.89$, $p < 0.05$) and positively correlated with evacuation time in Category 3–4 compound conditions ($r = 0.97$, $p < 0.01$), suggesting that higher unemployment is associated with both greater disruption-related delay and poorer evacuation outcomes when hazards compound.

Figure 7 shows the spatial distribution of social vulnerability (SoVI) scores across Westbrook at the block-group scale, classified into four levels (Low, Low–Medium, Medium–High, and High). Block Groups 1 and 2 fall in the Low category (0–0.25), whereas Block Groups 3, 4, and 5 fall in the High category (0.75–1.0). These higher-vulnerability block groups are concentrated along the coast and overlap with areas characterized by higher SLOSH surge exposure.

5.3. Model Optimization

Figure 8 illustrates the effect of optimizing evacuation behavior, defined as selecting the in-town shelter share that minimizes evacuation failure, across block groups and scenarios. The optimized in-town share varies by block group and storm condition, ranging from approximately 12% to 88%, reflecting differences in local accessibility and shelter constraints. In scenarios where evacuation failure is already zero, the baseline destination split is retained.

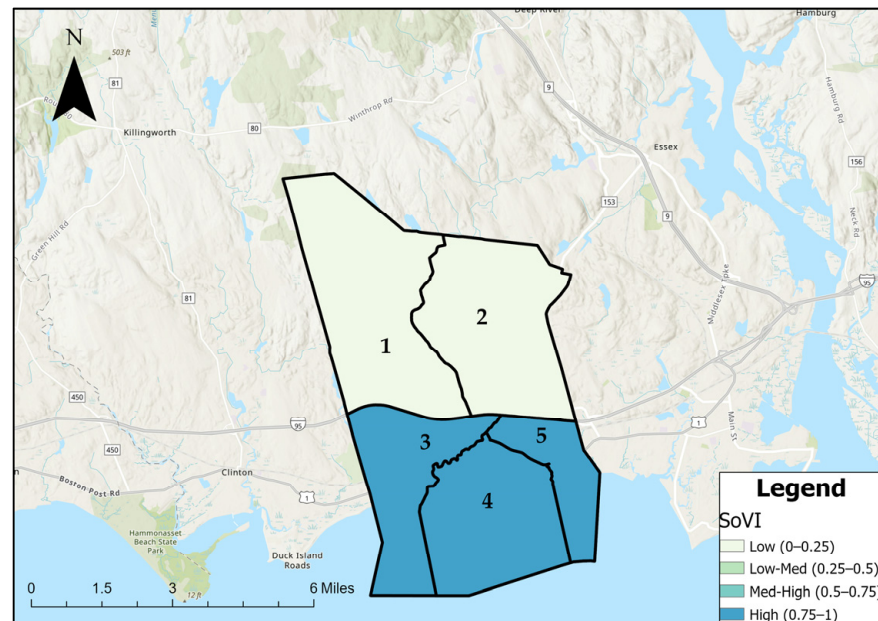


Figure 7. Spatial Distribution of Social Vulnerability Index (SoVI) Scores across Block Groups in Westbrook, Connecticut.

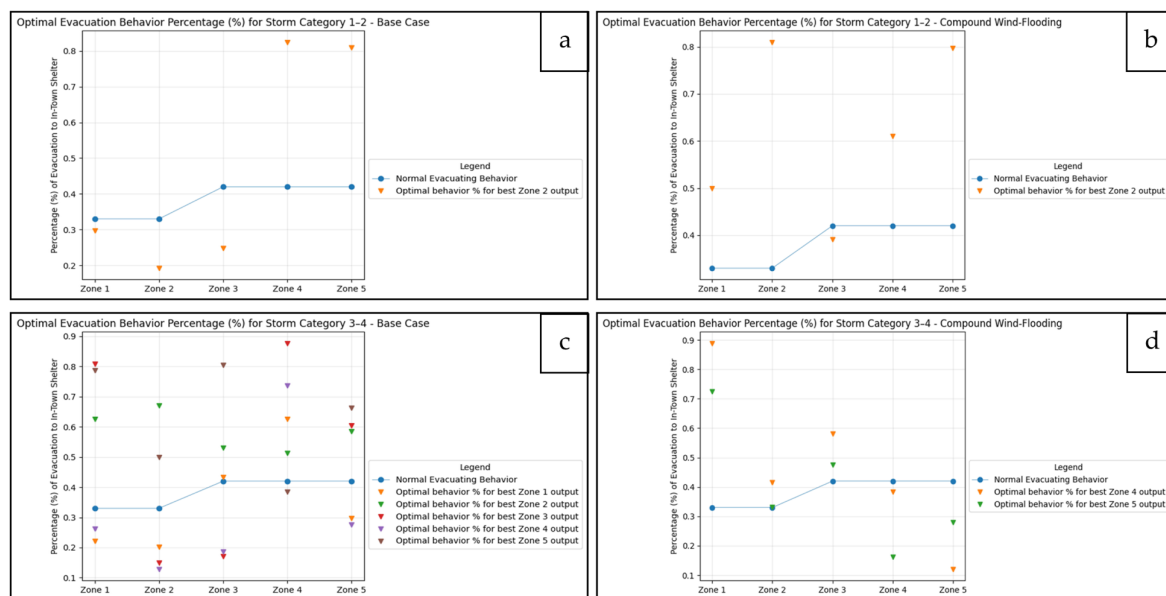


Figure 8. Optimized Evacuation Behavior and Resulting Evacuation Failure across Block Groups. (a) Category 1–2 Baseline, (b) Category 1–2 Compound Wind–Flooding, (c) Category 3–4 Baseline, and (d) Category 3–4 Compound Wind–Flooding Scenarios.

Table 4 summarizes the effect of optimizing evacuation behavior, defined as the share of evacuees selecting in-town shelters, on evacuation failure across block groups and scenarios. Under Category 1–2 baseline conditions, evacuation failure is already near zero in most block groups, and the optimized behavior produces only marginal changes. Minor reductions are observed in Block Group 2 under Category 1–2, where failure declines to approximately 0% in the baseline case and to about 3% under compound wind–flooding. In contrast, the largest benefits appear under Category 3–4 baseline conditions, where evacuation failure decreases in all block groups, with reductions of approximately 5–6% in Block Groups 1 and 2, 10.3% in Block Group 3, 16.6% in Block Group 4, and 30.5% in Block Group 5 relative to the original evacuation behavior. Under Category 3–4 compound wind–

flooding conditions, the optimized destination split yields only marginal benefits in the most affected areas. Block Groups 4 and 5 improve by approximately 1–2 percentage points, yet evacuation failure remains high (89–90%). This pattern suggests that destination-choice adjustments alone are unlikely to substantially reduce failure under severe compound disruption and that complementary measures, such as earlier departure timing, additional shelter capacity, or targeted improvements to critical links through redundancy and debris clearance, may be required to achieve meaningful performance gains in these block groups.

Table 4. Reduction in Evacuation Failure under Optimized Evacuation Behavior by Block Group and Scenario.

Storm Scenario	Evacuation Failure (%) (Reduction %)				
	Block Group 1	Block Group 2	Block Group 3	Block Group 4	Block Group 5
Category 1–2 (Baseline)	-	0 (−5.30%)	-	-	-
Category 3–4 (Baseline)	5.9% (−5.70%)	16.2% (−5.60%)	15% (−10.30%)	20.5% (−16.55%)	28.5% (−30.47%)
Category 1–2 (Wind–Flooding)	-	3% (−0.20%)	-	-	-
Category 3–4 (Wind–Flooding)	-	-	-	89.2% (−1.82%)	89.6% (−1.00%)

To evaluate the effect of increasing shelter capacity in a high-impact area, an additional shelter was introduced in Block Group 3, which exhibited the longest evacuation times as well as elevated failure under compound-hazard scenarios reaching approximately 90.7% (Figure 5d). The added shelter was assigned a capacity of 500 people and placed at the Westbrook Town Hall to support accessibility and operational feasibility during evacuation. The site was selected to avoid road segments subject to storm-surge inundation or debris-related closures in the modeled scenarios, ensuring that the added capacity remained accessible. This experiment was evaluated independently of behavioral optimization to isolate the effect of added capacity on evacuation time and failure.

Figure 9 shows the effect of the added shelter on evacuation time and evacuation failure in Block Group 3. Under Category 1–2 baseline conditions, evacuation failure remains at zero with or without the added shelter, while evacuation time decreases from approximately 3.0 h to 2.25 h. Under Category 3–4 baseline conditions, evacuation failure decreases from approximately 25.4% to zero after the shelter is added, while evacuation time remains approximately 3.6 h. Under Category 1–2 compound conditions, evacuation failure remains approximately 20.9%, but evacuation time decreases from about 4.0 h to 1.57 h. The largest effect occurs under the Category 3–4 compound scenario, where evacuation failure decreases from approximately 90.7% to 63.3% and evacuation time decreases from approximately 19.6 h to 3.26 h. Overall, added shelter capacity substantially improves performance in Block Group 3 under higher-intensity and compound-hazard scenarios, with the strongest gains observed when network disruption and shelter constraints coincide.

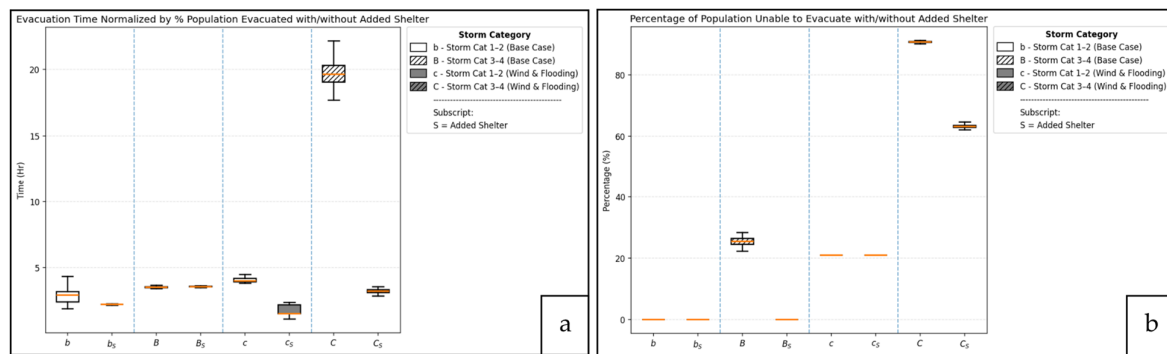


Figure 9. Effect of Additional Shelter Capacity in Block Group 3 on (a) Evacuation Time, and (b) Evacuation Failure.

6. Discussion

A strong spatial stratification in evacuation resilience across Westbrook becomes most pronounced under compound disruption. Under baseline conditions, the network remains broadly connected and performance is governed primarily by demand and geometric constraints, yielding relatively consistent normalized evacuation times. As storm intensity increases, outcomes diverge across block groups, implying that exposure to system stress is uneven and concentrated in neighborhoods whose egress depends on a limited set of vulnerable links or on narrow connectivity to shelters and inland routes. Under compound wind–flooding conditions, concurrent inundation and debris-related blockages reduce redundancy and shift the system from congestion-limited delay to connectivity-limited evacuation, where feasibility of reaching a safe destination becomes the dominant determinant of performance. The concentration of severe delay and failure in Block Groups 3–5 is consistent with this transition and suggests higher sensitivity to link loss and isolation.

An additional implication is that compound hazards can redistribute network demand in ways that complicate interpretation of aggregate travel-time measures. Shorter evacuation times in Block Groups 1 and 2 under Category 3–4 compound conditions reflect reduced network loading when highly exposed block groups contribute fewer trips due to disconnection, thereby lowering congestion and shelter queueing for connected areas. This “selective clearance” effect illustrates why town-wide averages can mask equity-relevant outcomes and reinforces the value of jointly evaluating evacuation time and success at the block-group scale.

Destination-specific patterns further suggest that out-of-town sheltering functions as a coupled regional capacity problem rather than an independent local option. High out-of-town success in baseline scenarios indicates adequate corridor access and capacity under moderate demand, whereas performance declines under severe and compound scenarios as regional shelter availability is reduced by concurrent demand in surrounding towns and by increased constraints along key connectors. Consequently, shifts toward out-of-town evacuation may yield limited benefit when capacity and accessibility are jointly stressed, highlighting the need for planning that treats shelter capacity, routing, and inter-town competition for resources as interdependent.

The socioeconomic results indicate that evacuation performance is spatially stratified in Westbrook and closely tracks the town’s vulnerability geography. Block Groups 1–2 fall in the low SoVI class, whereas Block Groups 3–5 fall in the high SoVI class, and this separation is mirrored in outcomes: high-SoVI block groups repeatedly exhibit longer normalized evacuation times and higher evacuation failure across storm categories, with disparities widening under compound wind–flood disruption. Interpreted jointly, the pattern suggests that unequal evacuation outcomes reflect not only differences in household resources and

constraints, but also the co-location of socially vulnerable populations with higher surge exposure and more fragile local connectivity, which amplifies the consequences of roadway loss and shelter saturation when hazards compound.

The correlation analysis provides complementary evidence that evacuation outcomes covary with indicators commonly discussed in the vulnerability and evacuation literature [113]. These relationships should be carefully interpreted as associations rather than causal effects, especially given the small number of block groups. The results are consistent with prior syntheses emphasizing that socioeconomic and demographic characteristics are linked to preparedness and to the willingness and means to evacuate [113].

Under baseline conditions, the positive association between Asian population share and in-town evacuation success is best read as descriptive of Westbrook's population geography and access conditions (e.g., proximity to shelters, and network connectivity), rather than as a standalone behavioral mechanism. In the Category 1–2 baseline scenario, the perfect positive association between two-parent-family prevalence and out-of-town evacuation success, and its perfect negative association with evacuation failure, align with the interpretation that two-caregiver households, particularly those with children, have stronger incentives and greater capacity to evacuate, given their ability to share childcare responsibilities and coordinate evacuation decisions and logistics. The perfect negative association between low educational attainment and out-of-town success, coupled with a perfect positive association with evacuation failure, is consistent with the broader framing that education and related resources are protective because they shape access to information, planning capacity, and feasible evacuation options. Under the Category 3–4 baseline scenario, the significant associations of limited English proficiency and lack of health insurance with longer evacuation times, and renter occupancy with higher evacuation failure and lower out-of-town success, align with a “means-and-constraints” perspective. Language barriers and unmet service needs can delay information uptake and increase planning friction, while renter-heavy contexts can signal lower resource buffers, weaker destination networks, and greater housing-related uncertainty, all of which reduce flexibility in departure timing, routing, and destination choice when demand is high. The significant association between female population share and lower evacuation time should be interpreted cautiously: although prior studies often report higher evacuation propensity among women, block-group-level correlations may instead reflect spatial confounding, where gender composition co-varies with hazard exposure, housing patterns, or transportation and network accessibility rather than directly shaping evacuation behavior.

In Category 1–2 compound conditions, the association of Social Security reliance with reduced out-of-town success and increased evacuation failure is consistent with the idea that fixed-income dependence can limit the ability to adapt plans as access and capacity degrade. In Category 3–4 compound conditions, the near-perfect positive association between unemployment and evacuation time, together with its positive association with evacuation failure in Category 1–2, suggests that economic insecurity is strongly aligned with prolonged clearance and lower attainment of safe destinations. Across scenarios, the stronger alignment between the social vulnerability assessment with evacuation performance metrics indicates that compound hazards can magnify pre-existing disparities, not by changing who is vulnerable, but by increasing the consequences of constraints that are already present.

Congestion results indicate that evacuation delay in Westbrook is governed by a small set of junctions that function as network control points, because they concentrate turning and merging movements and provide access to the primary shelter and egress corridors. This is evident in the repeated appearance of the same intersections across scenarios, particularly those involving U.S. Route 1 and Connecticut Route 153. For example, U.S.

Route 1 at Route 153 consistently operates as a convergence node for multiple block groups entering the main arterial system, while Route 153 at McVeagh Road concentrates inbound flows toward the largest-capacity in-town shelter. Similarly, Route 153 at Bokum Road emerges as a key outbound junction that channels traffic toward out-of-town sheltering and inland connections. In functional terms, these sites regulate clearance dynamics because queues formed there propagate upstream into neighborhood streets and downstream along arterial links, affecting travel times for multiple origins simultaneously.

Compound wind–flood disruption can alter evacuation dynamics by degrading network connectivity and reshaping the spatial distribution of evacuation demand. When inundation and debris-related blockages constrain egress from the most exposed block groups, fewer vehicles can access primary corridors. This can reduce observed queueing in the areas that remain connected, even as overall system performance worsens through higher evacuation failure elsewhere. This “selective loading” effect is important for interpretation: localized congestion relief should not be interpreted as improved efficiency but may instead indicate disconnection-driven reductions in participation from isolated neighborhoods. From a planning standpoint, the findings suggest that delay mitigation and failure reduction are related but distinct objectives. Managing delay emphasizes operational measures at critical junctions and connectors (e.g., traffic control, signal timing, turn management, and rapid incident clearance), whereas reducing evacuation failure requires maintaining viable paths to safe destinations for the most exposed neighborhoods through redundant egress options and robust access to shelters under compound disruption.

The optimization results indicate that destination guidance can reduce evacuation failure when the system is primarily constrained by shelter competition and corridor saturation, rather than by loss of connectivity. In the Category 3–4 baseline case, the large improvements in the higher-risk block groups imply that the survey-based destination split can concentrate demand on specific shelters and outbound routes, pushing them past operational thresholds and producing avoidable overflows. The fact that the optimal in-town share differs by block group underscores a planning implication: destination choice is a spatial demand-allocation mechanism, not a uniform townwide preference. Rebalancing the split redistributes flow across the coupled shelter–network system, easing bottlenecks and reducing the probability that evacuees arrive after capacity is exhausted. Under compound wind–flooding, the sharply reduced benefit reflects a regime shift in which link failures and isolation dominate outcomes; once feasible paths to shelters are removed, changing destination preferences cannot materially increase success because accessibility, not choice, becomes the binding constraint.

The added shelter improves performance through a different pathway by increasing reachable capacity within a high-impact origin area and reducing the travel burden under disruption. The marked decreases in failure and normalized evacuation time for Block Group 3, especially under Category 3–4 compound conditions, suggest that a substantial share of failure in that area was driven by capacity shortfalls among accessible shelters and the resulting longer, more failure-prone trips. The limited spillover benefit to Block Groups 4 and 5 clarifies why single-site capacity additions are unlikely to resolve townwide deficits under compound hazards: benefits remain localized unless the facility is accessible from multiple high-deficit origins under the disrupted topology and has sufficient capacity to absorb both local and redirected demand. Practically, these results support a portfolio approach in which shelter additions are paired with siting for multi-origin accessibility and with complementary measures that reduce demand peaks or restore connectivity on critical links.

These findings suggest that evacuation timing can deliver substantial system-wide benefits without changes to infrastructure or destination behavior. Using the baseline

clearance time as an operational benchmark links evacuation orders to measurable system performance and indicates when departures must begin to complete clearance before storm surge and debris reduce connectivity and usable shelter capacity. In the most constrained compound scenarios, early activation can reduce evacuation time by more than 15 h and avoid failure increases exceeding 60 percentage points in the highest-risk block groups. Overall, baseline clearance time offers a practical pre-event planning threshold for evacuation resilience, indicating how early action can preserve network connectivity and shelter access before compound hazards substantially degrade evacuation performance.

7. Conclusions and Future Work

7.1. Main Contribution and Findings

This paper presents a micro-scale, agent-based evacuation model for evaluating municipal evacuation performance under baseline conditions and under compound hurricane disruption from wind and flooding. Implemented in AnyLogic and demonstrated for the Town of Westbrook, Connecticut, the framework emphasizes interpretability and practical decision support by integrating empirically informed evacuation participation and destination tendencies from the New England Household Evacuation Survey, block-group demographic and vehicle availability from the American Community Survey, capacity-constrained shelter information from FEMA's National Shelter System, and a GIS-based road network representation. The approach produces spatially explicit, planning-relevant outputs, including evacuation time, shelter demand and unmet capacity, congestion locations, and evacuation failure, enabling fine-scale assessment of how performance varies across neighborhoods and storm conditions. The simulations reveal substantial spatial variation in evacuation performance across Westbrook and strong sensitivity to compound wind–flood disruptions. Under baseline conditions, evacuation times range from approximately 2.8–4.3 h for Category 1–2 storms and increase to 4.3–6.3 h for Category 3–4 storms, with the longest delays occurring in Block Groups 4 and 5. When compound wind–flood disruptions are introduced, evacuation times increase sharply in several locations, reaching 19.6 h in Block Group 3 under the Category 3–4 compound scenario, while Block Groups 4 and 5 exceed 8.0 h and 7.5 h, respectively. Correspondingly, evacuation success deteriorates in these high-vulnerability areas, with failure rates exceeding 90% in Block Groups 3–5, compared to near-zero failure under baseline conditions. At the system level, the effective town-wide clearance time increases from approximately 6.3 h in the Category 3–4 baseline scenario to nearly 19 h under compound hazards, reflecting the combined effects of network disruption, rerouting, and shelter accessibility constraints. Optimization of destination choice reduces evacuation failure by 10–30% in several block groups under baseline conditions but provides only marginal improvements under severe compound disruptions. It also tests scenario-based interventions to identify practical levers for reducing delay and failure under both baseline and compound conditions. For example, the addition of a single shelter in Block Group 3 reduces evacuation failure from 90.7% to 63.3% and decreases evacuation time from 19.6 h to 3.26 h, demonstrating the significant role of localized shelter capacity in mitigating evacuation failure under compound hazards. The analysis further links modeled outcomes to block-group social vulnerability indicators to evaluate distributional inequities in evacuation performance. Block groups with higher renter shares, unemployment, and limited English proficiency tend to experience longer evacuation times and higher failure rates, while variables such as two-parent household share are associated with higher out-of-town evacuation success. Beyond the case study, the contribution is a transferable and reproducible approach that demonstrates how commonly available data sources and commercially available software can be leveraged to support evacuation

timing, shelter planning, and hazard mitigation for small and mid-sized communities facing compound hurricane impacts.

7.2. Limitations and Future Research

Several limitations motivate future research. First, departures are synchronized and do not represent late evacuation, re-entry, or within-event behavioral adaptation, which may be consequential when warnings evolve, compliance varies, and households revise plans as conditions deteriorate. Second, shelter representation focuses on public shelters and does not explicitly model private accommodations, informal sheltering, or household-specific constraints that affect destination feasibility. Third, hazard impacts are simplified. Storm-surge effects are implemented using static maximum inundation extents rather than time-varying flood progression, and wind-driven debris is represented probabilistically without explicit modeling of clearance operations, restoration sequencing, or partial lane reductions. Fourth, behavioral parameters are survey-derived and may not fully capture local compliance, risk perception, social-network effects, or heterogeneity in departure timing across household types. Finally, validation is constrained by the availability of event-specific observations at the municipal scale. Future work should incorporate time-staged departure and compliance dynamics, mixed sheltering options, time-varying hazard progression, explicit debris clearance and recovery processes, and expanded validation using event-based traffic counts, shelter intake records, and post-event field reports. This analysis could also be extended through formal sensitivity or ablation experiments, potentially using machine learning, to systematically quantify the relative contribution of individual model components to evacuation performance. An additional direction involves incorporating electric vehicles (EVs) into evacuation simulations. As EV adoption increases, evacuation dynamics may be influenced by vehicle range constraints, charging availability, charging duration, and power-system disruptions during extreme events. Integrating EV-specific mobility behavior and charging infrastructure within evacuation ABMs would allow future studies to evaluate evacuation feasibility under electrified transportation scenarios and assess the resilience of charging networks during hurricane evacuations [114]. These extensions would strengthen both the operational fidelity of the model and its transferability for preparedness, response, and resilience planning in coastal communities.

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